Childhood Systemic Lupus Erythematosus

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Definition

Childhood Systemic Lupus Erythematosus (cSLE) is a chronic heterogeneous autoimmune disease associated with considerable morbidity and significant mortality. It is characterized by multisystem sequential or simultaneous organ involvement (predominant musculoskeletal, cutaneous, renal, hematologic and neuropsychiatric manifestations) and presence of anti-nuclear and other antibodies.

Clinical Manifestations and Classification Criteria for SLE

Clinical domains and criteria (scores)

Constituitional		
Fever		2
Neuropsychiatric		
Delirium		2
Psychosis		3
Seizure		5
Musculoskeletal		
Joint involvement		6
Hematologic		
Leucopenia		3
Thrombocytopenia		4
Autoimmune hemolysis		4
Renal		
Proteinuria >0.5 g/24 h		4
Renal biopsy Class II / V lupus nephritis		8
Renal biopsy Class III/IV lupus nephritis		10
Mucocutaneous		
Non scarring alopecia		2
Oral ulcers		2
Subacute cutaneous lupus		4
Acute cutaneous lupus		6
Serosal		
Pleural/pericardial effusion		5
Acute pericarditis		6

Immunologic domains and criteria

Antiphospholipid antibodies		2
Anti-cardiolipin antibodies/ anti-beta-2-glycoprotein antibodies/lupus anticoagulant		1
Complement proteins		
Low C3 or low C4		3
Low C3 and low C4		4
SLE-specific antibodies		
Anti-dsDNA antibody or anti-Smith antibody		6

Total score of ≥10 and ≥1 clinical criterion AND ANA at a titer of ≥1:80 on HEp-2 cells required to classify SLE

Source: Adapted from European League Against Rheumatism/American College of Rheumatology classification criteria for systemic lupus erythematosus; 2019.

Severity of Illness

Mild SLE: No renal or other life-threatening organ system involvement

Moderate SLE: Clinically significant but not life-threatening involvement of kidneys or other vital organs/systems

Severe SLE: Major organ threatening disease

Investigations

Laboratory investigations form integral part of diagnosis and management of patients with SLE.

General laboratory investigations		
Erythrocyte sedimentation rate (ESR)	Increase in ESR is in proportion to disease activity. May be falsely elevated in hypergammaglobulinemia	
C-reactive protein	☑ Usually normal even with active disease ☑ Increased in patients with concomitant infection, serositis or arthritis	
Ferritin	Increased, albeit not as much as in macrophage activation syndrome	
Specific mmunological investigations		
	Method of detection	***Comments***
Antinuclear antibodies (ANA)	Indirect immunofluorescence on HEp2 cells remains the gold standard for detection	Present in more than 90% of children with lupus. Patterns of staining may reflect antigenic specificity of the antibody
Anti-double stranded antibodies (anti-dsDNA)	Enzyme immunoassay or immunofluorescence	☑ High specificity for SLE ☑ Marker for active disease especially lupus nephritis
Complement estimation	Nephelometry or enzyme immunoassay	Complement C3 and C4 are commonly measured. Low levels detected in active disease
Extractable nuclear antigens	Can be detected by immunoblotting or by counterimmunoelectrophoresis	Provide useful information on antigenic specificity of ANA
Antiphospholipid (aPL) antibodies	☑ ELISA for anticardiolipin and anti-β2-glycoprotein I antibodies ☑ Lupus anticoagulant assay by activated partial thromboplastin time (aPTT) and dilute Russell's viper venom time (dRVVT)	Positive in significant proportion of pediatric SLE patients. Anticoagulants (heparin and oral anticoagulants) may interfere with detection of lupus anticoagulant

Treatment

Childhood systemic lupus erythematosus ranges in severity from mild disease with arthritis and rash to devastating nephritis with renal failure or profound neurological disturbances. Approach to therapy is individualized for each patient and based on the clinical features. Nevertheless, treatment is unified based on a few guiding principles.

Ten Principles of Treatment

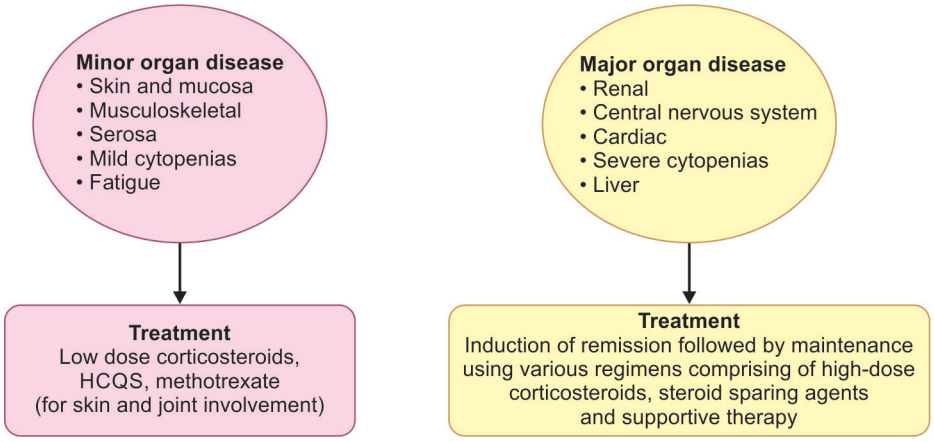
1. Treatment by multidisciplinary team with the paediatric rheumatologist as the Lead
2. Assess organ involvement
3. Assess disease activity and damage index
4. Direct therapy to specific disease manifestations that are active
5. Achieve remission with adequate immunosuppression
6. Therapy for the shortest time period and at lowest possible doses
7. Maintain remission with minimal toxicity of drugs
8. Prompt recognition and aggressive treatment of 'flares' and infection
9. Counsel and educate parents and child
10. Transition to adult care in late adolescence

General Treatment

- Adequate nutrition
- Avoid sun exposure, use water-based sunscreen with SPF >30
- Aggressive treatment of infections
- Immunize for age with non-live vaccines in cSLE (influenza, hepatitis A and B, human papilloma, meningococcal, pneumococcal, COVID-19)
- Calcium and vitamin D supplementation if on steroids folic acid supplementation if on methotrexate
- Recognise "Flare of disease activity" (development of new symptoms and signs which require a change in therapy)
- Maintain normal blood pressure
- Maintain attendance and participation in all school activities to maximum extent possible
- Ensure good mental health

Organ Specific Treatment

Based on severity of involvement



Assess Disease Activity and Damage Index

Assessment of disease activity and damage is best done by pediatric rheumatologist using SLE Disease Activity Index (SLEDAI)/British Isles Lupus Assessment Group (BILAG) index and paediatric version of American College of Rheumatology/Systemic Lupus International Collaborating Clinics (ACR/SLICC) damage score index.

Counseling and Education

- ☑ About chronic nature of illness with flares and remissions
- ☑ Regular follow-up
- ☑ Monitoring for side effects of drugs
- ☑ Need for prolonged therapy
- ☑ Access to patient information material
- ☑ Need for shared care with multidisciplinary team
- ☑ Career choices
- ☑ Transition to medical care in late adolescence
- ☑ Coping strategies for parents and child

Name	Dosage	Indications	Important side effects	Monitoring
Prednisolone	*Initiation of therapy* (4–6 weeks): 0.5–2 mg/kg/day (max 60 mg/day) in at least two divided doses based on type and severity of organ involved. *Tapering dose*: Reduce by 2.5–5 mg/week till a daily dose of 20 mg Further reduction by 1–2.5 mg every 2–4 weeks till a daily dose of 10 mg After a daily dose of 10 mg is reached, reduce by 0.5–1 mg every 2–4 weeks	Mainstay of treatment to induce and maintain remission	Obesity, glaucoma, cataract, hypertension, growth suppression, diabetes, osteoporosis, avascular necrosis of bone, infections	Regular blood pressure and random blood sugar monitoring, calcium and vitamin D supplement, bone density monitoring
Intravenous pulse methyl prednisolone	10–30 mg/kg to a maximum of 1 g/day for 1–5 consecutive days followed by oral prednisolone as above	Severe cytopenias, proliferative nephritis, major CNS disease, vasculitis, "lupus crisis"	Hypertension, hyperglycemia Rest of the side-effects similar to prednisolone	Monitoring of vitals especially blood pressure before starting and during infusion
Hydroxy-chloroquine	4–6 mg/kg/day at bedtime (maximum dose 400 mg/day)	Cutaneous disease and adjunct to steroids for systemic disease	Nausea, pruritis, rarely maculopathy	Eye check at starting drug and biannually for visual acuity, color vision testing, visual field testing, and retinoscopy
Methotrexate	10–15 mg/m^2 subcutaneously weekly once	☑ Arthritis ☑ Resistant skin disease	GI intolerance, mouth ulcers, alopecia, bone marrow suppression, hepatotoxicity	AST/ALT, serum albumin, CBC at baseline and then at 8–12 weeks intervals

Contd...

Contd...

Name	***Dosage***	***Indications***	***Important side effects***	***Monitoring***
Nonsteroidal anti-inflammatory drugs (NSAIDs)			GI intolerance/ bleed, renal impairment, hypersensitivity	AST/ALT, serum creatinine at baseline and 6 month intervals
Naproxen	15–20 mg/kg/day in two divided doses (maximum dose 1 g)	Myalgia, arthralgia, arthritis	Pseudoporphyria-pruritis, urticarial, morbiliform rash, erythema multiforme	
Ibuprofen	30–40 mg/kg/day in three divided doses (maximum dose 2.4 g)	Positive aPL antibodies		
Aspirin	3–5 mg/kg/day			
Cyclophos-phamide	1–2 mg/kg/day (PO) or 500–1000 mg/m^2/month (IV) in severe disease	Severe cytopenias, proliferative nephritis, major CNS disease, vasculitis	Nausea, vomiting, cytopenias, infections, hemorrhagic cystitis	Ensure adequate fluid intake CBC on day 7 and 14 if on pulse therapy
Azathioprine	0.5–2.5 mg/kg/day (PO) (Maximum 150 mg/day)	Mild SLE unresponsive to glucocorticoids and HCQS or develop unacceptable drug toxicity	Cytopenias, pancreatitis Opportunistic infections	CBC monthly and after dose increments, hepatic enzymes, BUN, creatinine amylase and lipase at onset of treatment and every 3 months
Mycophenolate Mofetil	1200–1800 mg/m^2/day twice daily (PO) (Maximum 3000 mg/day)	Induction and maintenance of severe cytopenias, vasculitis, proliferative and membranous nephritis, steroid sparing in moderate to severe illness	☑ GI intolerance ☑ Pancytopenia ☑ Opportunistic infections	CBC every 8–12 weeks

Contd...

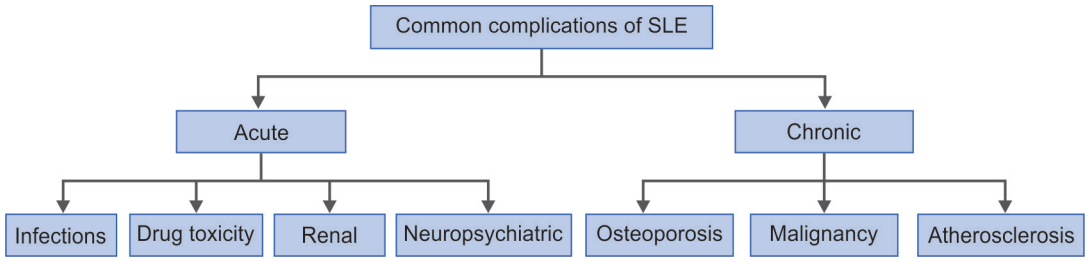
Drugs used in Management of SLE

Contd...

Name	Dosage	Indications	Important side effects	Monitoring
Cyclosporine A	2.5–5 mg/kg/day	Membranous nephropathy	nephrotoxic	Blood urea nitrogen, creatinine, urine analysis, hepatic enzymes, CBC at start of therapy and monthly
IV Immuno-globulin	Up to 2 g/kg	Severe thrombocytopenia, catastrophic anti-phospholipid syndrome	Anaphylaxis, Aseptic meningitis (Headache, vomiting 18–36 hours after the infusion)	Monitor blood pressure and pulse rate every 15 minutes for first hour, every 30 minutes for second hour, every hour thereafter
Rituximab	375 mg/m^2/week intravenously (four doses at interval of 7 days) OR 750 mg/m^2 (two doses at interval of 14 days)	Severe and refractory disease	Infection Hypogamma-globulinemia Infusion reaction	Monitor CBC at start of therapy
Belimumab (approved for children > 5 years)	10 mg/kg every 4 weeks	Clinically active disease without severe nephritis or neuropsychiatric lupus	Infection	Monitor CBC at start of therapy and monthly

(CBC: complete blood count; ALT: alanine transaminase; AST: aspartate aminotransferase)

Complications



Role of General Pediatrician (GP)

- Early recognition and referral to specialist (URGENT referral if major organ involvement)
- Liaison with rheumatologist for shared care
- Treat intercurrent infections aggressively
- Never stop immunosuppression without discussion with the rheumatologist
- Maintain immunization schedule. Administer non-live vaccines. Discuss about live vaccines with the rheumatologist
- Counsel parents and child

Prognosis

Early diagnosis and appropriate therapy ensures good prognosis. Intercurrent infections, flares, renal and cardiovascular disease, neurologic manifestations and poor adherence to treatment affects outcome. Morbidity of the disease and adverse effects of the medications must be minimized to achieve satisfactory long-term outcome.

Neonatal Lupus

- Misnomer as the neonate is not affected by SLE and in addition, the mother is often asymptomatic or suffering from an autoimmune disease other than lupus
- *Etiology:* Transplacental transfer of maternal autoantibodies with narrow range of antigenic specificity
- Autoantibodies are usually specific for Ro/SSA antigen, La/SSB antigen or rarely U1RNP
- Mere transfer of maternal autoantibodies is not sufficient for the development of neonatal lupus
- Congenital heart block is a dreaded complication of neonatal lupus

Drug-induced Lupus

Medications associated with drug-induced lupus

- ☑ Procainamide
- ☑ Hydralazine
- ☑ Minocycline
- ☑ Isoniazid, Rifampicin
- ☑ Phenytoin
- ☑ Carbamazepine
- ☑ Valproate
- ☑ Penicillamine
- ☑ Amiodarone

Clinical characteristics

- ☑ Males and females equally affected
- ☑ Hepatitis more common
- ☑ Anti-dsDNA and hypocomplementemia less common
- ☑ Significant renal/neurologic disease less likely
- ☑ Anti-histone antibodies (>95 %)

Management

- ☑ Withdraw the offending medication
- ☑ Hydroxychloroquine, NSAIDs and steroids may be required for short duration

Lupus Mimics

Infections	
Viral	Epstein-Barr virus, Cytomegalovirus, Parvovirus B19, Human Herpes Virus 6
Bacterial	Brucellosis, Leptospirosis, Q fever, Mycoplasma
Spirochetal	Lyme disease
Protozoal	Toxoplasmosis
Malignancy	Leukemia, Lymphoma
Other autoimmune diseases	Mixed connective tissue disease, systemic vasculitis, antiphospholipid syndrome, juvenile dermatomyositis

Summary

Diagnosis and management of cSLE is a challenge due to its myriad clinical presentations. Shared care pathways between specialists from different disciplines and different levels of care (primary, secondary, and tertiary) are of paramount importance in management of cSLE.

Further Reading

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Henoch-Schönlein Purpura

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Henoch-Schönlein Purpura

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Definition

IgA vasculitis (IgAV), formerly known as Henoch-Schönlein purpura (HSP), is an immune complex-mediated small vessel vasculitis of childhood. It is the most common pediatric vasculitis affecting children of all age groups.

TABLE 1: Classification criteria for IgA vasculitis.

Criteria	***Definition***
Purpura (mandatory)	Purpura (palpable, in crops) or petechiae, with lower limb predominance, not related to thrombocytopenia
And at least one out of four of the following:	
Abdominal pain	Diffuse, acute, colicky pain
Arthritis, arthralgias	*Arthritis:* Acute joint swelling or pain with limitation of motion *Arthralgia:* Acute joint pain without joint swelling or limitation of motion
Renal involvement	*Proteinuria:* >0.3 g/24 hr; spot urine albumin to creatinine ratio >30 mmol/mg; or ≥2+ on dipstick *Hematuria:* Red cell casts; urine sediment showing >5 red cells per high-power field or red cell casts
Histopathology	Leukocytoclastic vasculitis with predominant IgA deposits (skin biopsy); or proliferative glomerulonephritis with predominant IgA deposits (kidney biopsy)

Source: Adapted from Ozen, et al. EULAR/PRINTO/PRES criteria for Henoch-Schönlein purpura.